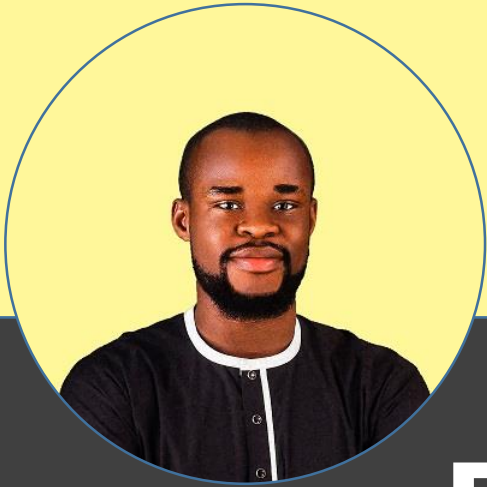




Business Intelligence Analytics

My Process and how I answered business questions and provided valuable and actionable insights



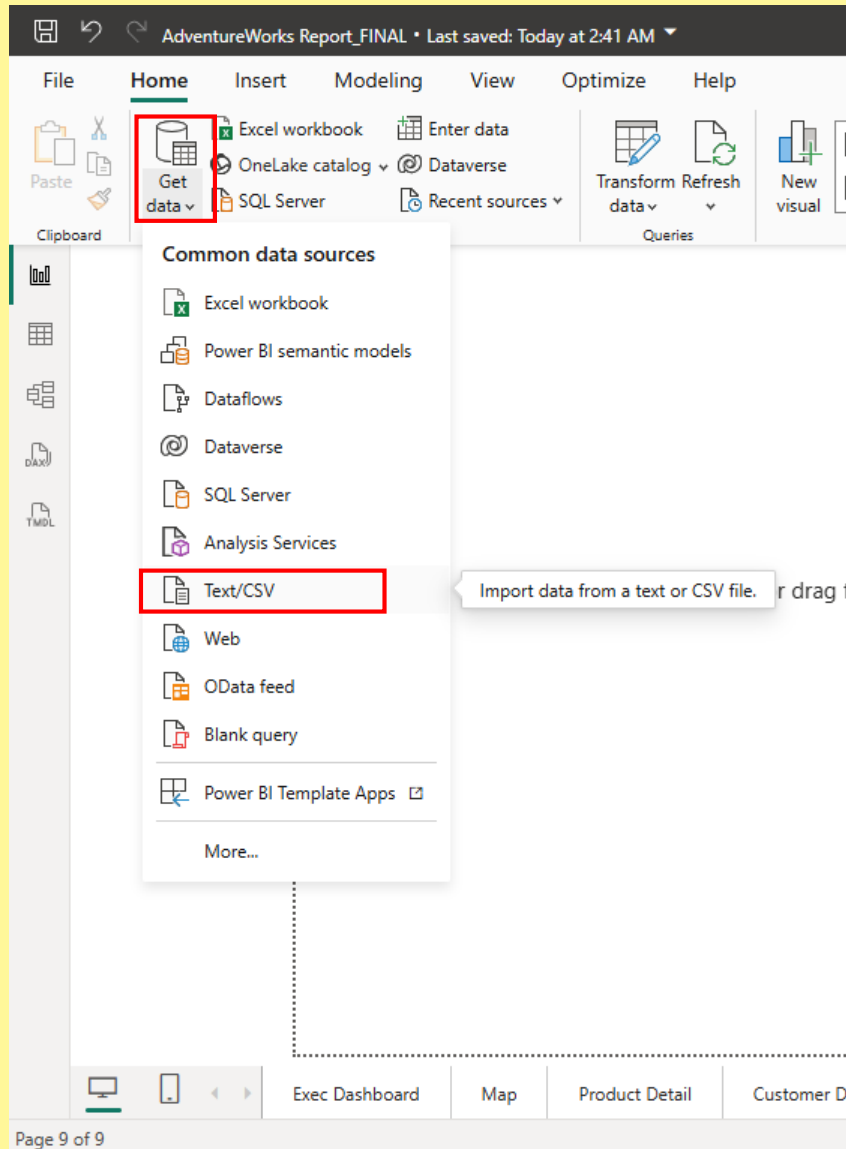
Overview

This project involved developing an interactive Power BI dashboard to analyze sales performance and key business KPIs. By consolidating revenue, profit, orders, customer, product, and regional data into a single reporting solution, the dashboard enables stakeholders to monitor performance, identify trends, and make informed, data-driven decisions.

Business Problems/Questions

1. What is the business performance based on the KPI (Revenue, Profit, Orders return Rate)?
2. Which Customer has invested the most over the years?
3. Which region has produced the most revenue?
4. What products are the most sold and the most returned? And why?
5. What category of products has the most orders?
6. Predict revenue with a 10% increase in price.
7. How is each product performing?
8. What are the top 10 products by orders?
9. What is the revenue trend over the years?
10. What is the revenue, profit and returns performances compared with previous months
11. What is the average revenue generated per customer?
12. What is the total order by Income level and occupation?

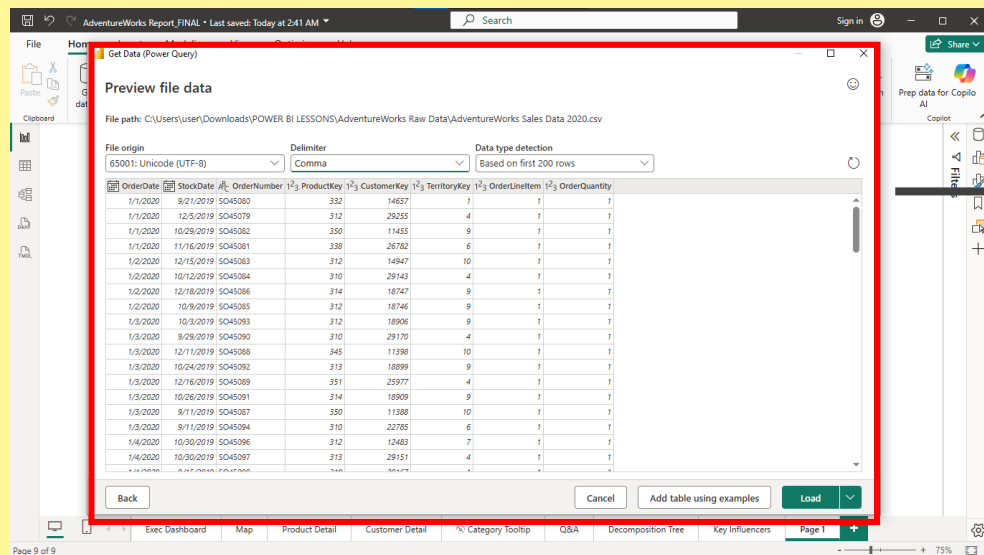
1. Importing the Files



There are a number of ways to get your data into the work space including from databases, but for the sake of the demonstration we will be importing Text/CSV files already extracted from the data base using SQL.

Clicking on the get data icon brings a pop up that prompts you to choose import style and then locate the file on your computer

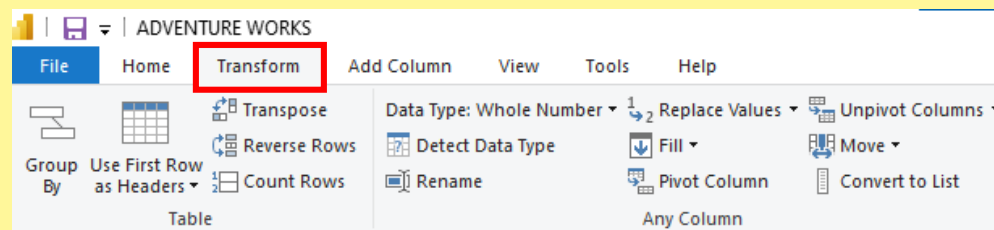
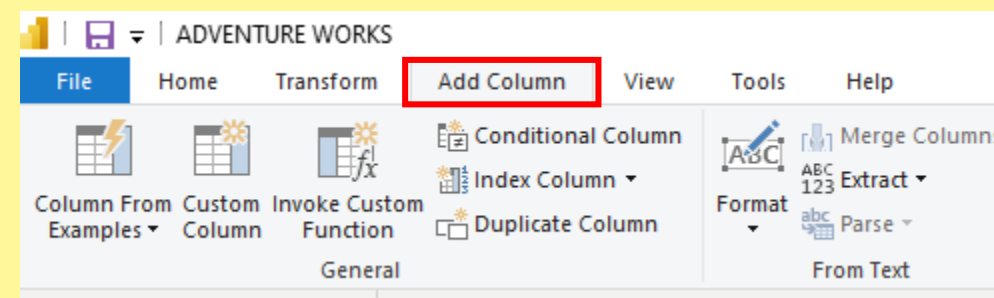
2. Data Cleaning (Power Query)



After selecting your data cleaning is important is this happens through the Power Query, a powerful tool that allows irregular, missing values to be tackled and also allows the creation of calculated columns.

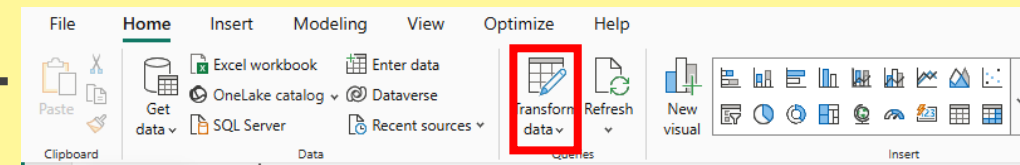
The add column and transform tabs and parts of the power query tool that allows you modify your data before creating relationships. Following the need of the business there is usually a clear idea on tables that need to be included

N/B: these are great tools but it is important not to overuse them cause they can make the file really heavy creating a lot new and redundant data. DAX measures are a preferred way. So calculated columns should be used only when necessary

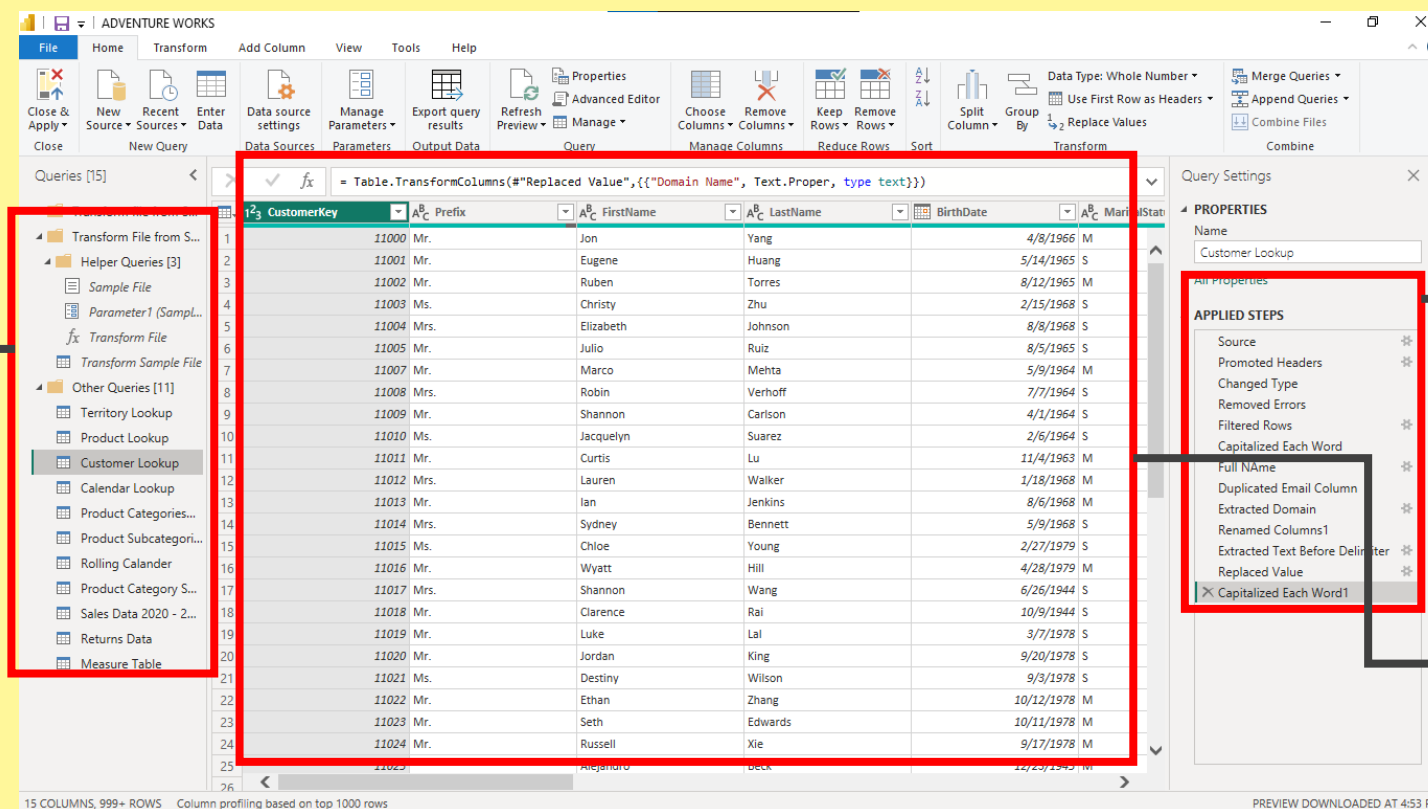


3. Previewing Data Tables after cleaning

After cleaning the data its important to carefully observe and walk through in case anything was missed. The Transform data tab allows you access the power query



This tab shoes all the tables loaded into the power query



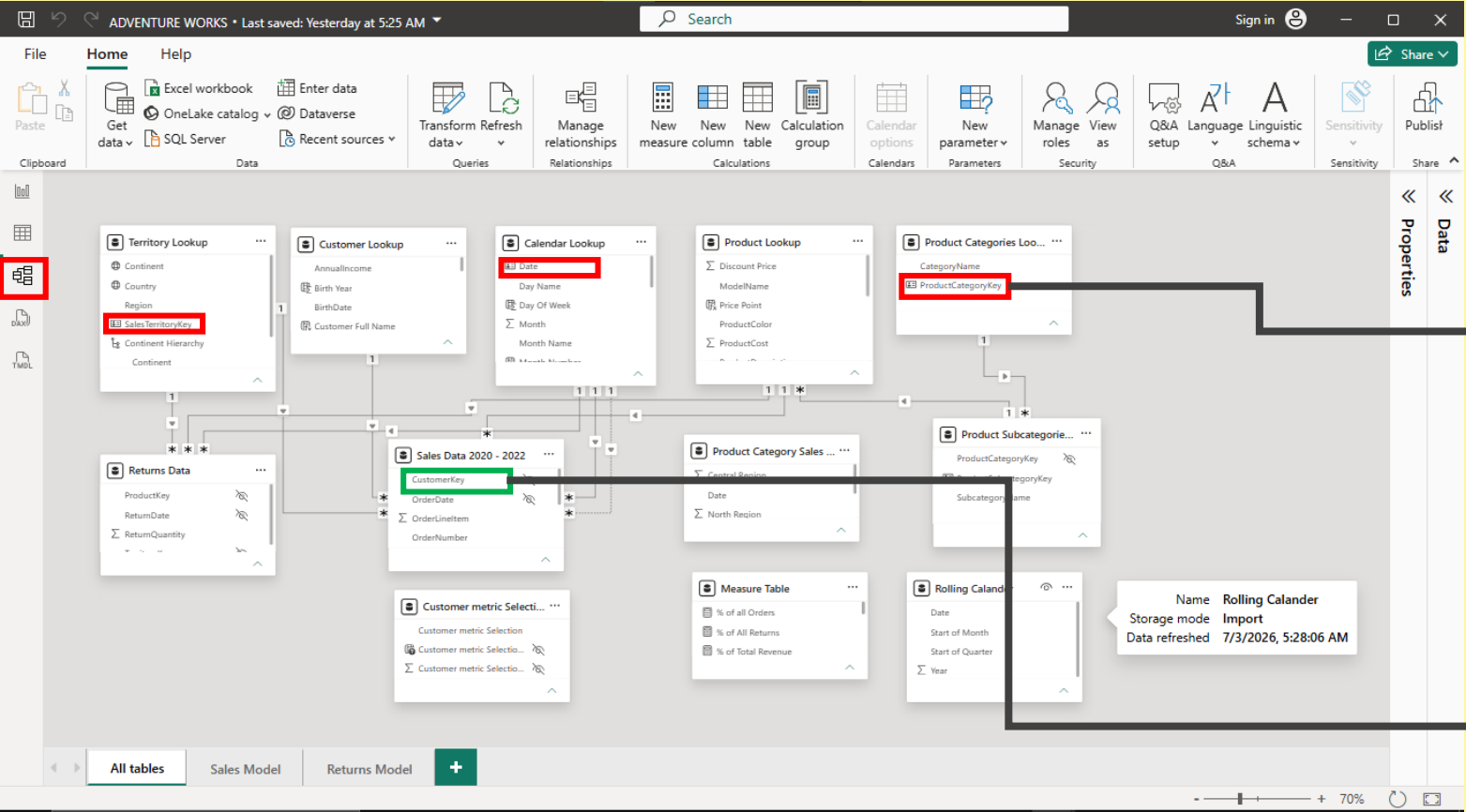
The applied steps tab shows all steps taken in transforming the data within the power query

This tab displays the current selected table

4. Data Modeling

This is a vital part of the analysis it involves creating relationships between previously independent tables allowing them to communicate without merging or combining them. This relationships are created through **Primary and Foreign keys**.

This tab opens the data modelling view



Columns in red are Primary keys which are unique identifiers of a table

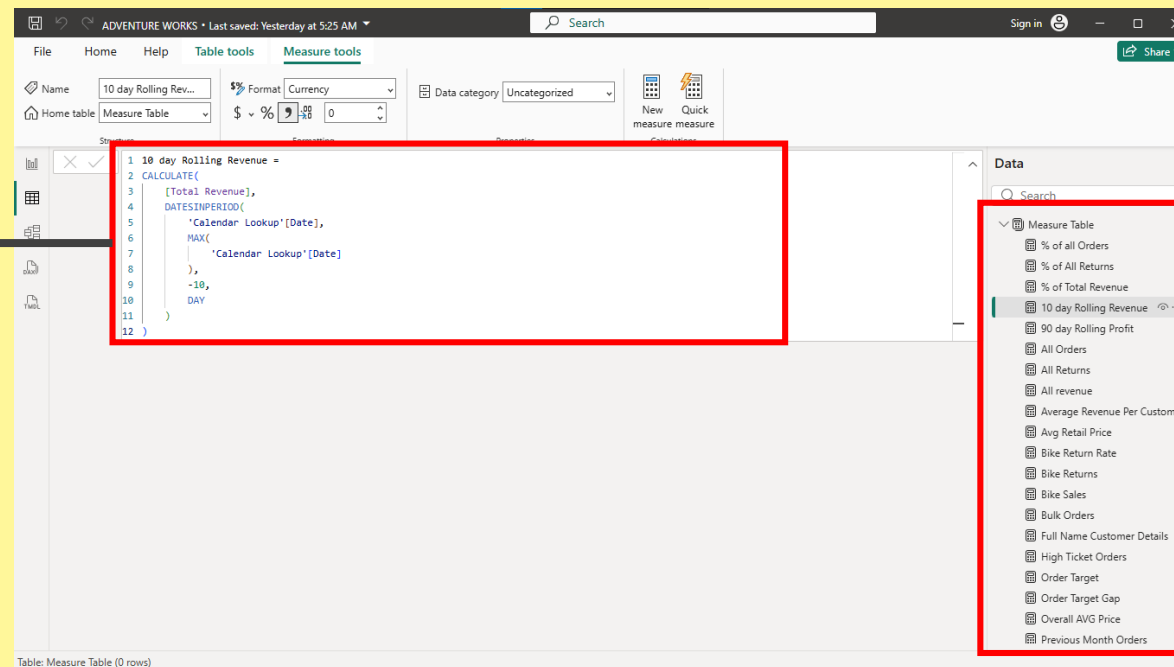
This is a foreign key it is a key that doesn't have unique entities

N/B: The pattern for connect a primary key to a foreign key is “one to many”

5. DAX

DAX measures are one of the amazing tools that allows a lot of computations and calculations without the need to create redundant calculated columns. They are a big part of the analysis process.

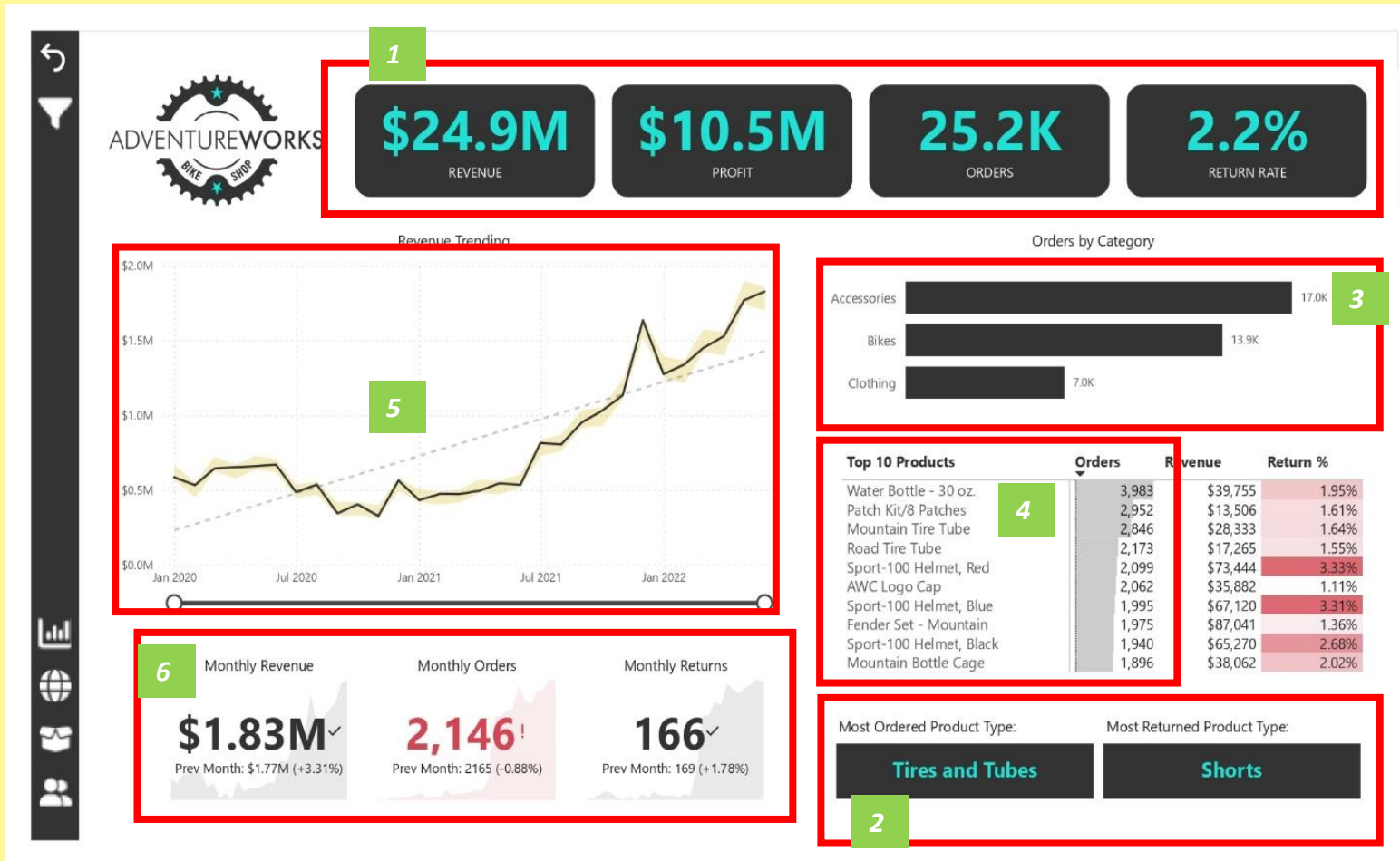
This showcases a sample of a dax measure created to calculate a 10 day Rolling revenue



This shows a table created to house all the DAX measures created in the course of this analysis

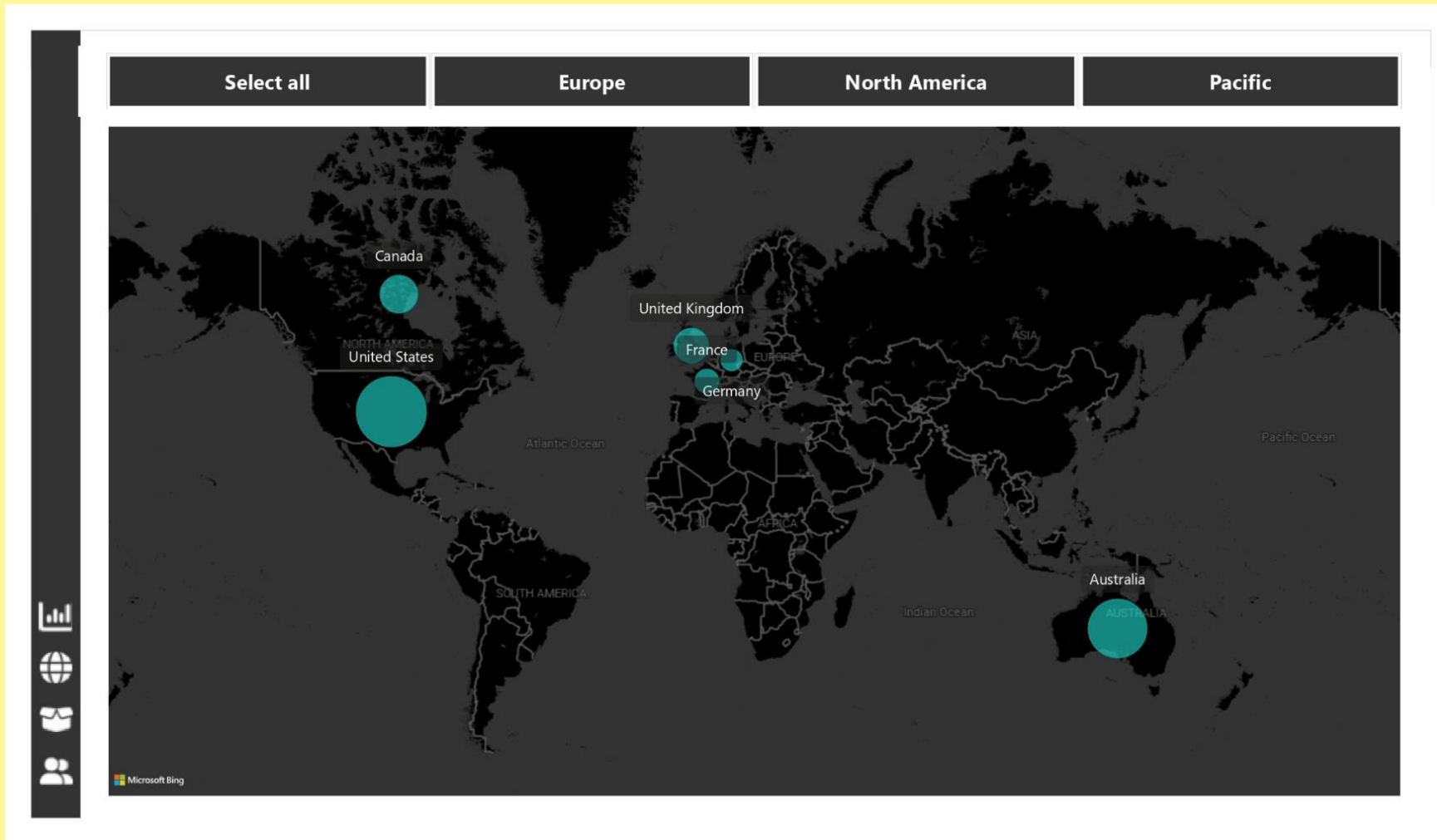
**Now we answer the business
questions using Data Visualization
(EDA)**

6. Data Visualization



1. What is the business performance based on the KPI (Revenue, Profit, Orders, Return Rate)?
2. What products are the most sold and the most returned? And why?
3. What category of products has the most orders?
4. What are the top 10 products by orders?
5. What is the revenue trend over the years?
6. What is the revenue, profit and returns performances compared with previous months

Data Visualization

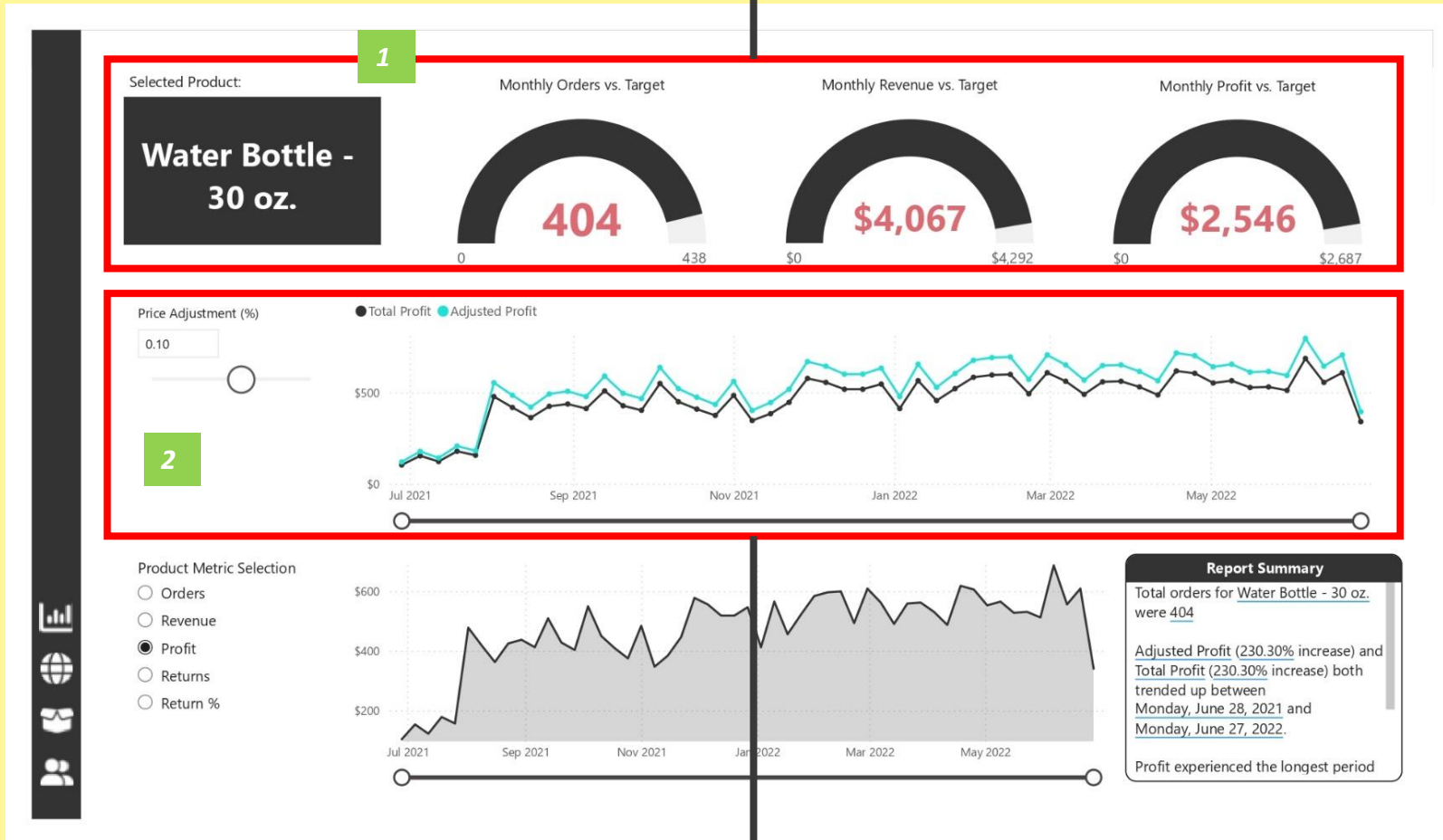


1. Which region has produced the most revenue?

The bubble size represents order quantity

Data Visualization

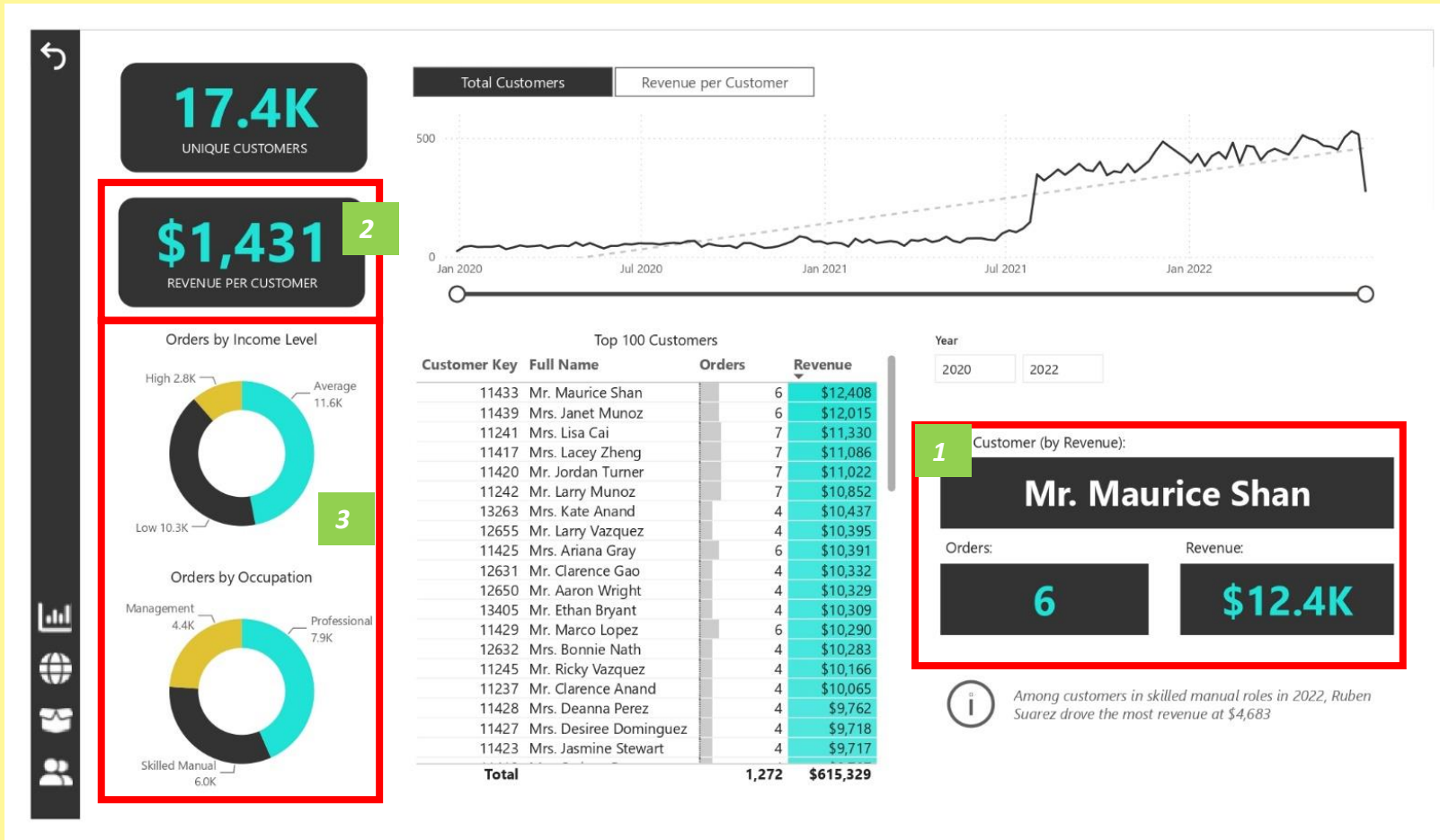
This shows the comparison of orders, profits and revenue between currently monthly amount and target amounts of each product.



1. Predict revenue with a 10% increase in price.
2. How is each product performing?

Slider on the far left allows us to view a predicted profit mount if prices were adjusted by a certain percent

Data Visualization



1. Which Customer has invested the most over the years?
2. What is the average revenue generated per customer?
3. What is the total order by Income level and occupation?

Conclusion

- I've heard people talk about the power of DAX measures, I didn't quite understand why,
- but doing this project, I made use of it 90% of the queries. • The range of flexibility it allowed, making possible the exploration of data in ways that calculated columns would struggle at.